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(.venv) @mehulyadav06 → /workspaces/ICP-0AEA6888-2026-REPO (main) $ /workspaces/ICP-0AEA6888-2026-REPO/.venv/bin/python /workspaces/ICP-0AEA6888-2026-REPO/Week3/chatbot.py
https://github.com/google-gemini/deprecated-generative-ai-python/blob/main/README.md

import google.generativeai as genai
=== Gemini Chatbot ===
Type 'exit' to quit.

You: Hello
Bot: Hello! How can I help you today?
You: What is AI?
Bot: AI stands for Artificial Intelligence.

In its simplest terms, AI refers to the simulation of human intelligence in machines that are programmed to think and learn like humans. It's about creating systems that can perform tasks that typically require human cognitive abilities.

Here's a breakdown of what that means:

1. Simulating Human Intelligence: AI aims to replicate human cognitive functions such as:
  * Learning: Acquiring information and rules for using the information.
  * Reasoning: Using rules to reach approximate or definite conclusions.
  * Problem-solving: Finding solutions to complex challenges.
  * Perception: Interpreting sensory information (like vision and hearing).
  * Language Understanding: Processing and generating human language.

2. How it Works (Generally): AI systems learn from data, identify patterns, and then make decisions or predictions based on those patterns. This learning often happens through algorithms, which are sets of instructions that the AI follows.

3. Types of AI:
  * Narrow AI (or Weak AI): This is the kind of AI we have today. It's designed and trained for a specific task. Examples include:
    * Voice assistants (Siri, Alexa)
    * Recommendation engines (Netflix, Amazon)
    * Spam filters in your email
    * Image recognition for facial identification
    * Self-driving cars (focused on driving)
    * Medical diagnosis tools
  * General AI (or Strong AI / AGI): This is hypothetical AI that would possess human-level cognitive abilities across a wide range of tasks, just like a human. It could learn, understand, and apply intelligence to any intellectual task. We are not yet close to achieving AGI.
  * Superintelligence: This is a hypothetical AI that would surpass human intelligence across virtually all fields, including scientific creativity, general wisdom, and social skills. Even further off than AGI.

4. Key Subfields of AI:
  * Machine Learning (ML): A subset of AI that focuses on enabling systems to learn from data without being explicitly programmed. It's the most common way AI is implemented today.
  * Deep Learning (DL): A subset of Machine Learning that uses neural networks with many layers ("deep" networks) to learn complex patterns from large amounts of data.
  * Natural Language Processing (NLP): Enables computers to understand, interpret, and generate human language.
  * Computer Vision (CV): Allows computers to "see" and interpret visual information from images and videos.
  * Robotics: Often integrates AI to make robots more autonomous and intelligent.

In essence, AI is about making machines intelligent, enabling them to automate complex tasks, discover insights, and interact with the world in more sophisticated ways. It's a rapidly evolving field that is transforming almost every industry.
You: Thank Your Response is very Helpful
Bot: You're most welcome! I'm glad I could help clarify what AI is.

If you have any more questions about AI, its applications, or anything else, feel free to ask! I'm here to assist.
You: exit
Bot: Goodbye!
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